CNS, Eye and Ear Infections

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Acknowledge Dr Stone's formatting and slides that have been edited by Dr Jensen with the help of Chat GPT for Clinicians –GPT-5.6 Sol and updated .

Objectives

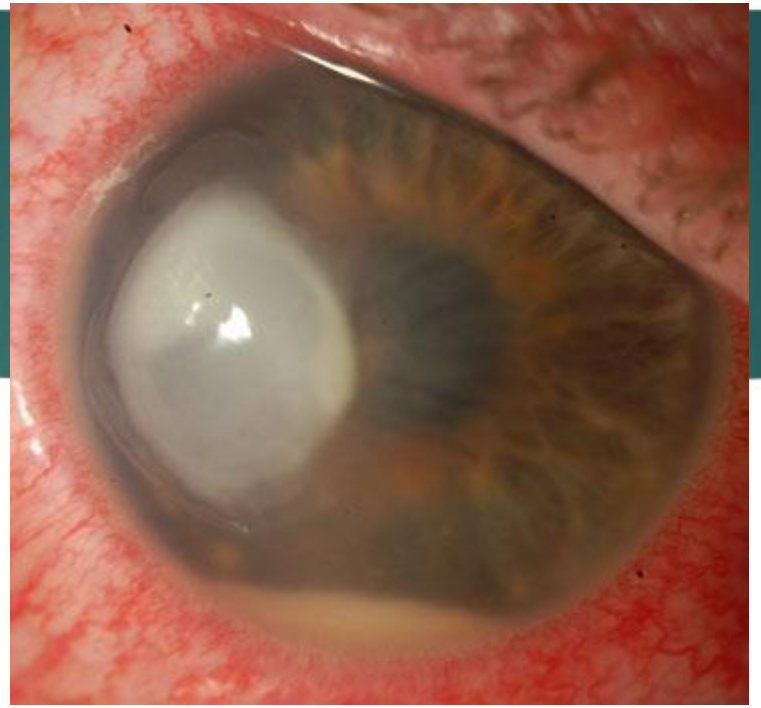
1. Differentiate the pathophysiology, clinical presentation, CSF findings, and diagnostic evaluation of common CNS infections. (CLO 3)

2. Correlate the routes of infection, causative microorganisms, and characteristic clinical features of bacterial, viral, fungal, and parasitic infections of the CNS. (CLO 3)

3. Correlate the clinical presentation and causative organisms of common infectious disorders of the eye and ear. (CLO 3)

Eye Infections

Corneal Ulcers



Clinical Presentation

Hallmark: Severe pain, tearing, and visible white or gray corneal lesion

Symptoms: Photophobia, discharge, possible vision impairment

Signs of tissue breakdown or necrosis

Pathophysiology

Deep tissue involvement of the corneal stroma

Often follows untreated keratitis or corneal trauma

Immune cells infiltrate and degrade corneal tissue, leading to ulcer formation

Etiological Agents

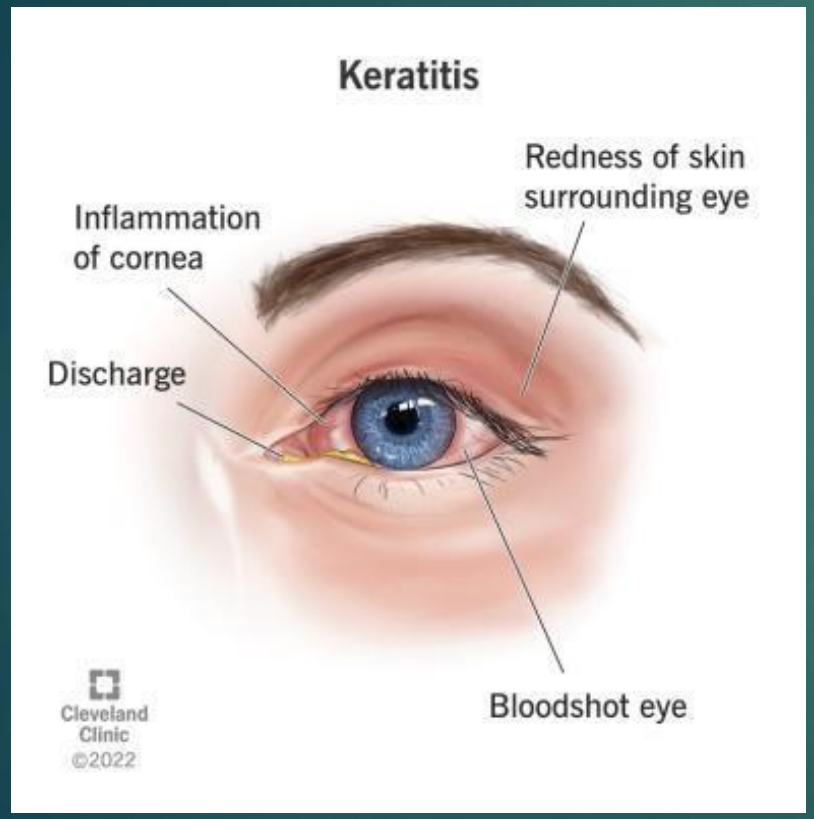
Baacterial: Staphylococcus species, Strep pneumoniae, Pseudomonas

Virus: Herpes simplex virus

Fungal: Aspergillus, Fusarium

Parasitic: Acanthamoeba in high-risk individuals (e.g., contact lens users)

Keratitis



Clinical Presentation

- Hallmark: Pain, light sensitivity, decreased visual acuity
- Symptoms: Foreign body sensation, redness around cornea
- Possible corneal opacities or infiltrates in infectious cases

Pathophysiology

- Inflammation and potential ulceration of the corneal tissue
- Pathogen-induced breakdown of corneal epithelium, leading to immune response
- Risk of scarring and vision loss if untreated

Etiological Agents

- Bacterial: Pseudomonas aeruginosa, Staphylococcus aureus, Streptococcus sp.
- Viral: Herpes simplex virus (HSV)
- Fungal and Parasitic Causes: Candida spp., Aspergillus spp., Acanthamoeba (contact lens wearers)

Conjunctivitis

Clinical Presentation

- Hallmark: Redness, "pink eye" appearance
- Symptoms: Tearing, irritation, possible discharge (purulent in bacterial, watery in viral)
- Commonly affects both eyes in infectious cases

Pathophysiology

- Inflammation of the conjunctiva
- Triggered by pathogens leading to immune cell infiltration
- Edema and hyperemia contribute to redness and discomfort

Etiological Agents

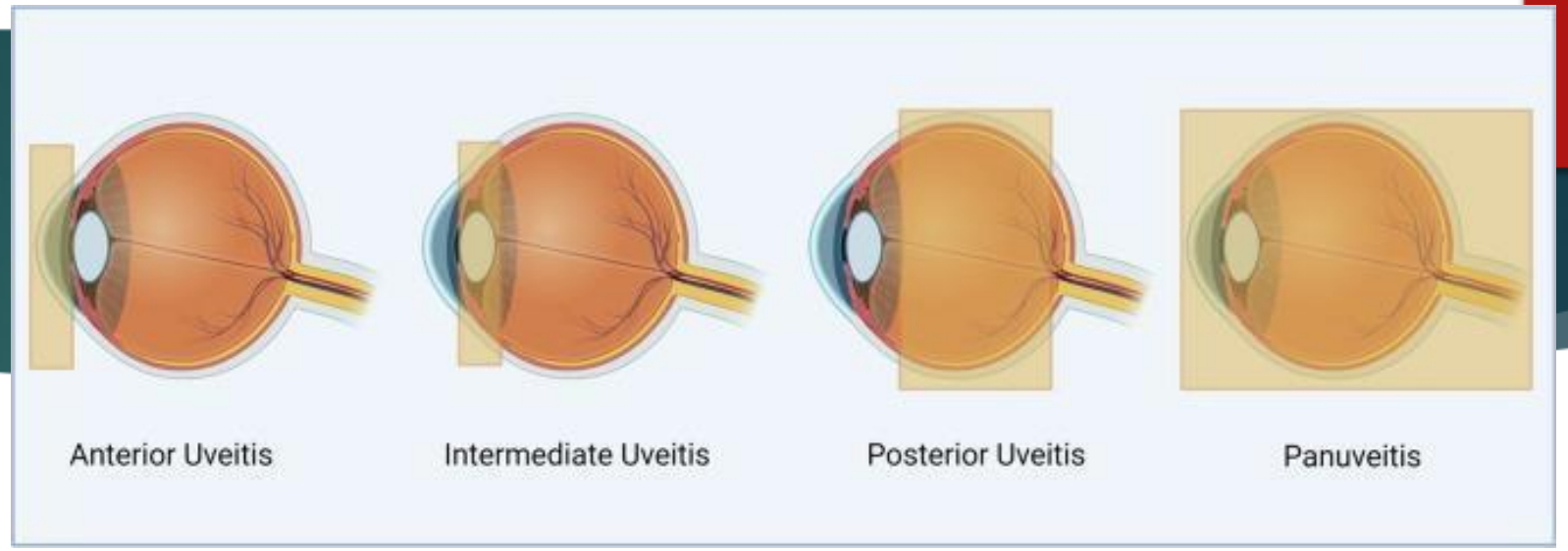
- Bacterial: Staphylococcus aureus, Streptococcus pneumoniae, Haemophilus influenzae, Chlamydia trachomatis (trachoma), Corynebacterium diphtheriae
- Viral: Adenovirus (most common)
- Allergic and Irritant Causes

Eye Discharge in Conjunctivitis



Mucopurulent discharge and eyelid matting increase the likelihood of bacterial conjunctivitis, but no single clinical finding reliably distinguishes bacterial from viral conjunctivitis."

Uveitis



Clinical Presentation

- Hallmark: Pain, photophobia, "floaters," blurred vision
- Symptoms: Redness concentrated around the limbus, decreased vision
- Unilateral or bilateral depending on cause

Pathophysiology

- Inflammation of the uveal tract (iris, ciliary body, choroid)
- Immune response against intraocular tissues, often with immune cell infiltration
- Can be granulomatous or non-granulomatous based on etiology

Selected infectious causes

- Herpes simplex virus (anterior), cytomegalovirus (posterior)
- Toxoplasma (posterior)
- Systemic Autoimmune diseases

Immune-mediated and idiopathic causes are major categories of uveitis.

Retinitis

Clinical Presentation

- Hallmark: Floaters, vision loss, scotomas
- Symptoms: Flashes of light, blurred vision
- Possible retinal hemorrhages or necrosis on fundoscopy

Pathophysiology

- Inflammation and damage of retinal tissue
- Often follows viral infections with direct cytopathic effects on retinal cells
- Can progress rapidly, leading to tissue necrosis and potential blindness

Etiological Agents

- Viral: Cytomegalovirus (especially in immunocompromised individuals)
- Fungal and Protozoan Causes: Toxoplasma gondii (rare)

Ear Infections

Acute Otitis Externa (AOE) – Clinical Presentation

Key Symptoms:

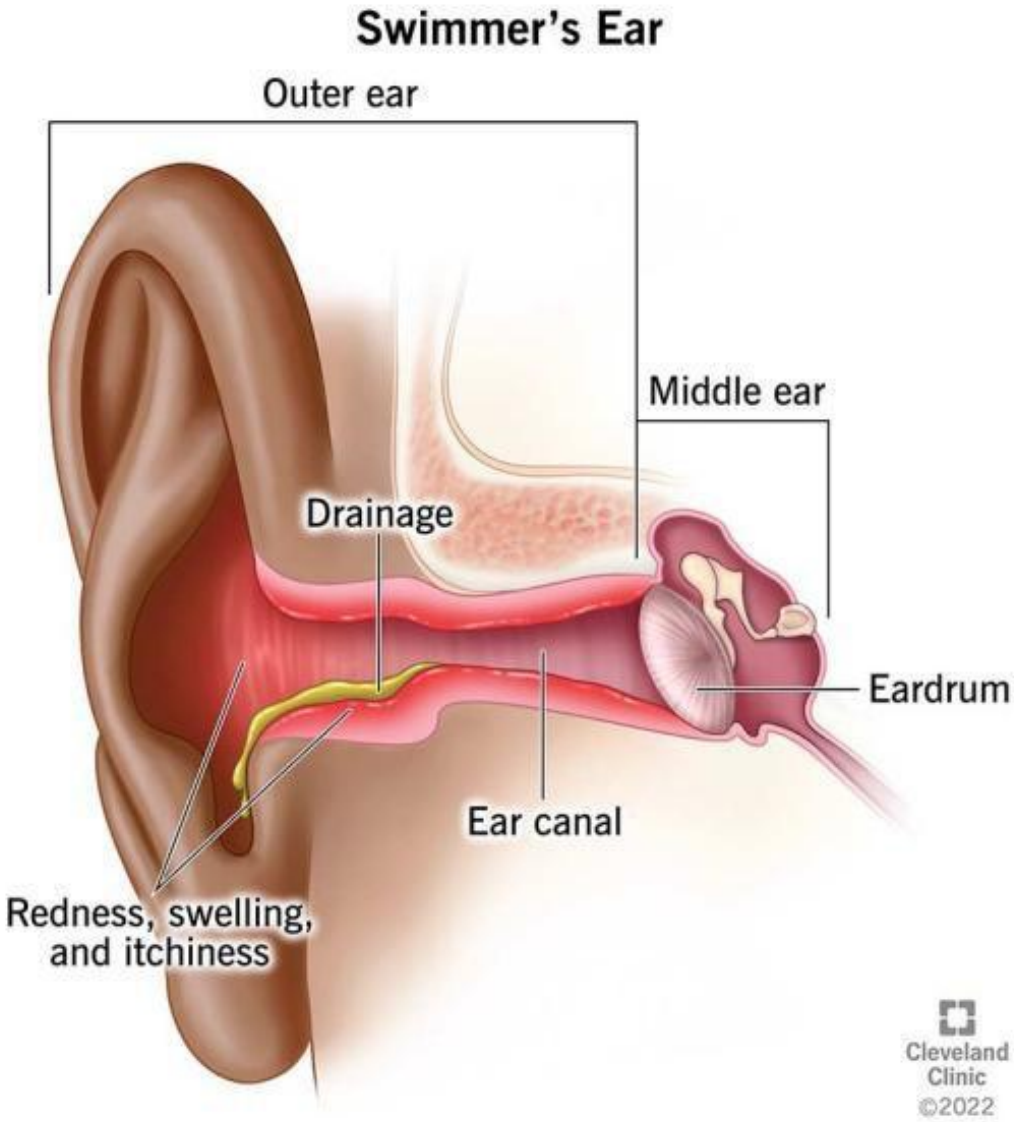
- Ear Canal Pain: Severe and often exacerbated by manipulation of the auricle or tragus.
- Itching: Common early symptom, especially in fungal cases.
- Discharge (Otorrhea): Clear, purulent, or even foul-smelling fluid may drain from the canal.
- Hearing Impact: Conductive hearing loss in severe cases due to obstruction.

Examination Findings:

- Canal Redness and Swelling: Often extends throughout the external auditory canal.
- Tenderness on Manipulation: Unique feature; pain increases with pulling on the ear.
- Possible Debris: Thickened discharge or fungal debris ("white, cottony" for fungal).

Differentiating Hallmarks:

- Pain upon ear manipulation and visible canal inflammation.
- Absence of TM bulging (unless complicated by middle ear involvement).



Acute Otitis Externa (AOE) – Pathophysiology & Etiology

Pathophysiology:

- Moisture Retention: Trapped moisture creates a favorable environment for microbial growth.
- Disruption of Skin Barrier: Minor trauma (e.g., from Q-tips or hearing aids) can predispose.
- Infection Progression: Pathogens colonize, leading to inflammation of the external canal.

Most Common Etiological Agents:

- Bacterial Causes:
  - Pseudomonas aeruginosa: Commonly found in water-exposed environments (e.g., swimming).
  - Staphylococcus aureus: Can contribute, especially with disrupted skin barriers.
- Fungal Causes:
  - Aspergillus spp. and Candida spp. seen more in chronic AOE or in immunocompromised patients.

Risk Factors:

- Frequent water exposure (e.g., swimming), excessive ear cleaning, use of hearing aids, eczema in the ear canal.

Acute Otitis Media (AOM) – Clinical Presentation

Key Symptoms:

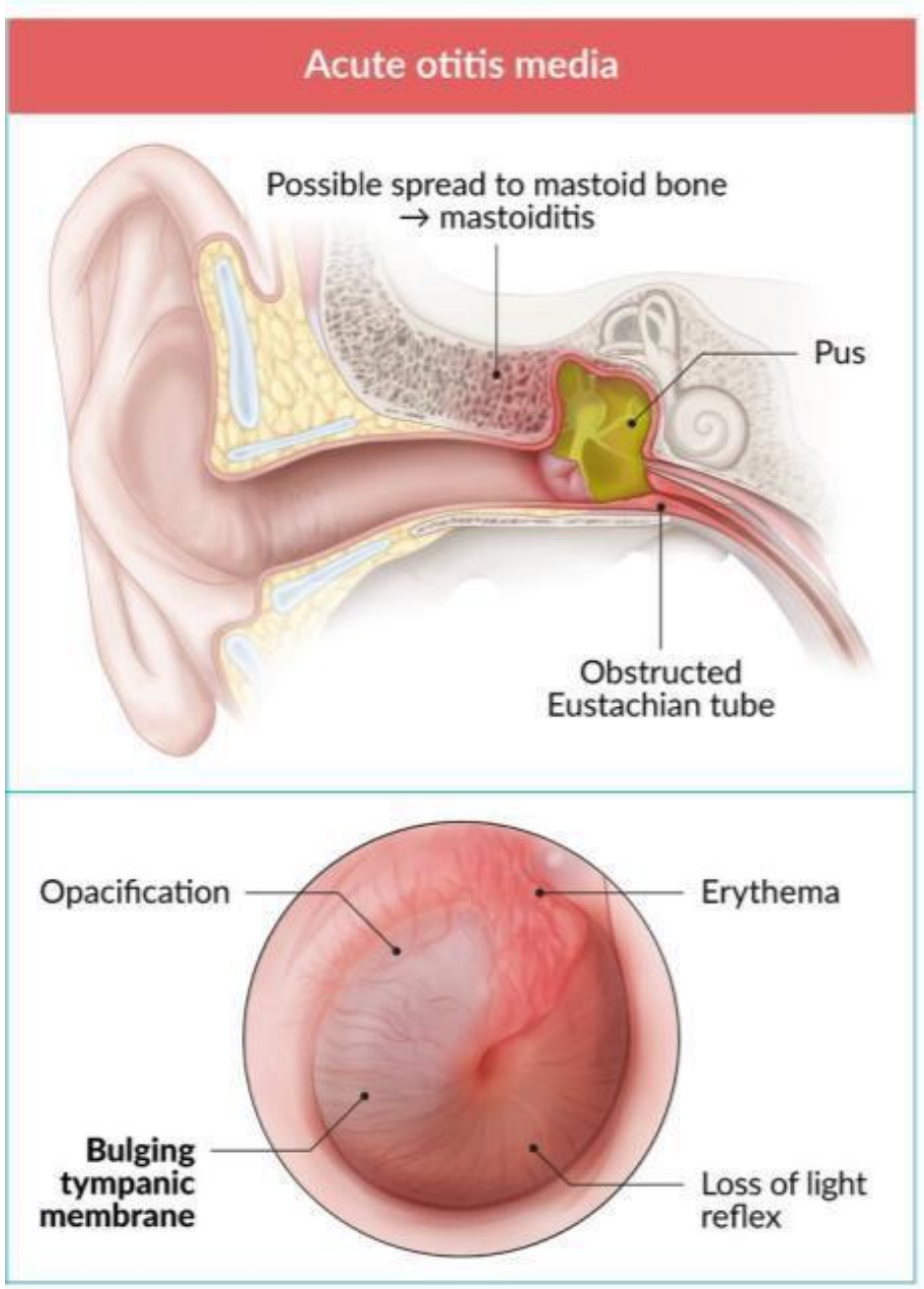
- Ear Pain (Otalgia): Deep, persistent pain often described as "fullness" in the ear.
- Hearing Loss: Due to fluid accumulation in the middle ear.
- Systemic Symptoms: Fever, irritability (more common in children), difficulty sleeping.

Examination Findings:

- Bulging Tympanic Membrane (TM): Characteristic bulging or opacity.
- Decreased TM Mobility: Assessed with pneumatic otoscopy, indicating fluid buildup.
- Redness: Inflammation visible on the TM surface.

Differentiating Hallmarks:

- Middle ear involvement with TM changes.
- Absence of external ear canal tenderness or pain with movement.



Acute Otitis Media (AOM) – Pathophysiology & Etiology

Pathophysiology:

- Eustachian Tube Dysfunction: Blocked tube leads to fluid accumulation in the middle ear.
- Fluid Stasis: Creates an environment conducive to bacterial or viral growth.
- Pressure Build-Up: Leads to TM bulging and associated pain.

Most Common Etiological Agents:

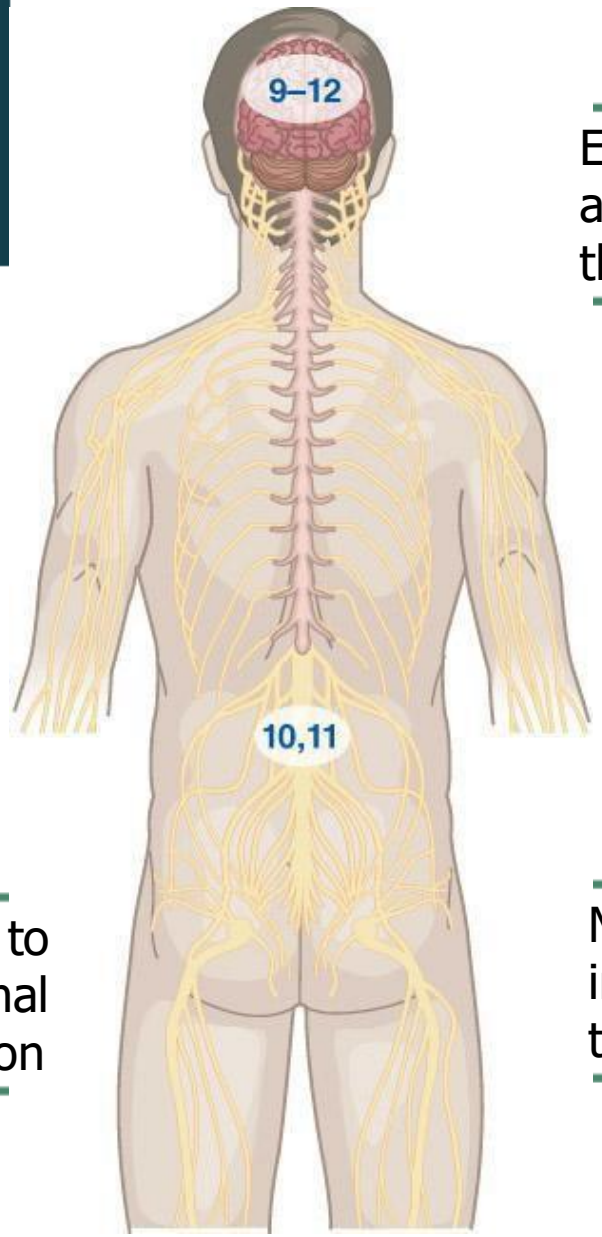
- Bacterial Causes:
  - Haemophilus influenzae: THE leading cause-non-typeable strains
  - Moraxella catarrhalis: Common in pediatric AOM.
  - Streptococcus pneumoniae: was the leading bacterial cause.
- Viral Contributors:
  - Respiratory syncytial virus (RSV), rhinovirus, and adenovirus may predispose to bacterial AOM.

Risk Factors

- Young age (eustachian tube anatomy), recent upper respiratory infections, allergies

CNS Infections

Areas of CNS damage



Encephalitis – infection and inflammation of the brain

Post-infection encephalitis* – The immune system attacks healthy brain tissue following an infection (days to weeks)

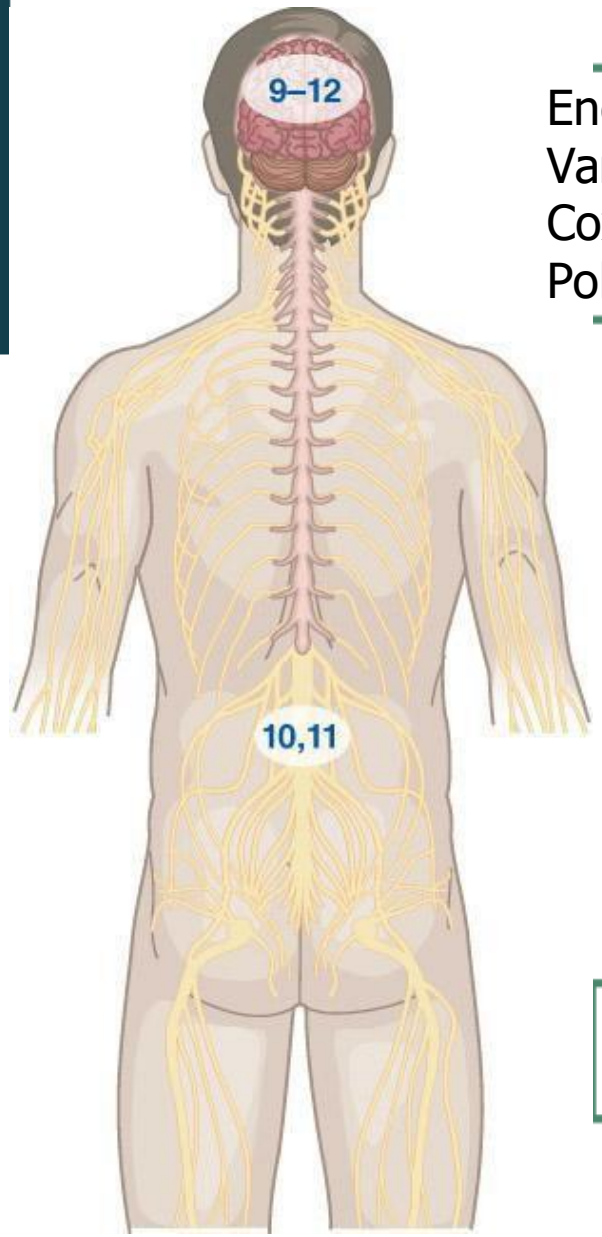
Meningitis – infection and inflammation of the meninges

Paralysis/Myelitis – Damage to the peripheral nerves or spinal cord leading to loss of function

Meningoencephalitis – infection and inflammation of the brain and the meninges

*aka Secondary encephalitis, Acute disseminated encephalomyelitis (ADEM) – Autoimmune mechanisms

Aseptic (Viral) causes of CNS damage



Encephalitis – HSV-1, Varicella-Zoster, Rabies, Coxsackieviruses A & B, Poliovirus

Post-infection encephalitis – Measles, Mumps, Rubella, Varicella-Zoster, Influenza

Meningitis – Enteroviruses (Echoviruses, Coxsackieviruses, Poliovirus), HSV-2, Adenoviruses, Mumps virus, Lymphocytic choriomenigitis virus (LCMV), Parainfluenza Types 2 & 3

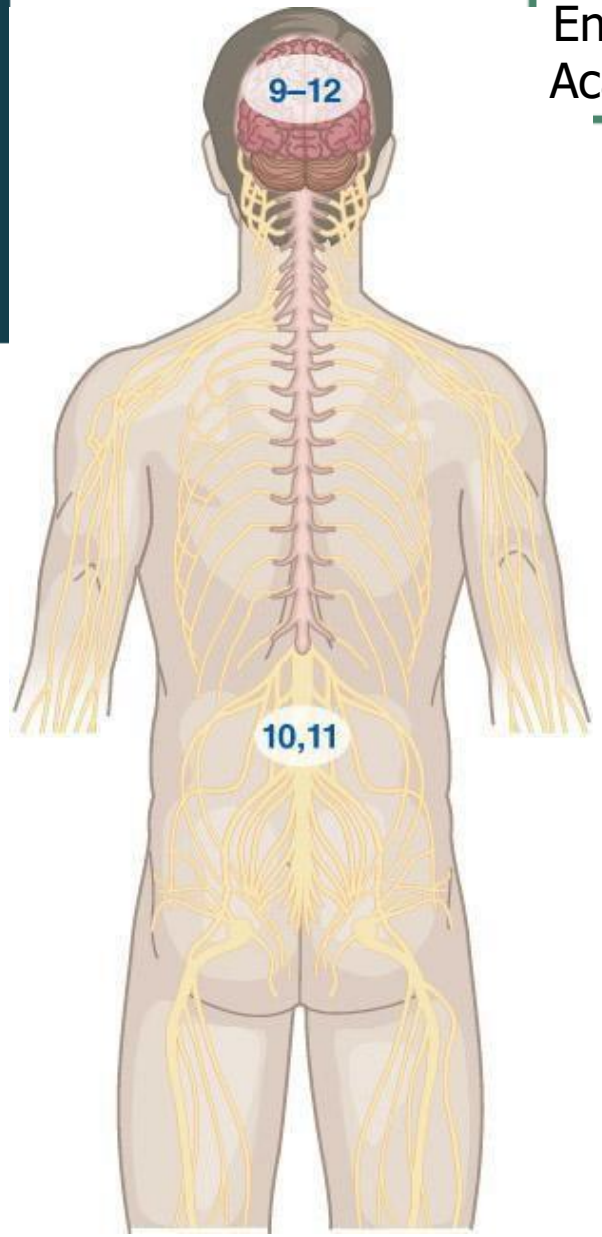
Paralysis/Myelitis – Poliovirus, Enteroviruses 70 & 71, Coxsackievirus A7

Meningoencephalitis – Arboencephalitis viruses, HIV

Septic causes of CNS damage

Meningitis (Acute) – E. coli, Group B streptococci, Neisseria meningitides, S. pneumoniae, Listeria monocytogenes

Meningitis (Chronic) – Mycobacterium tuberculosis



Encephalitis – Balamuthia, Acanthamoeba, Naegleria

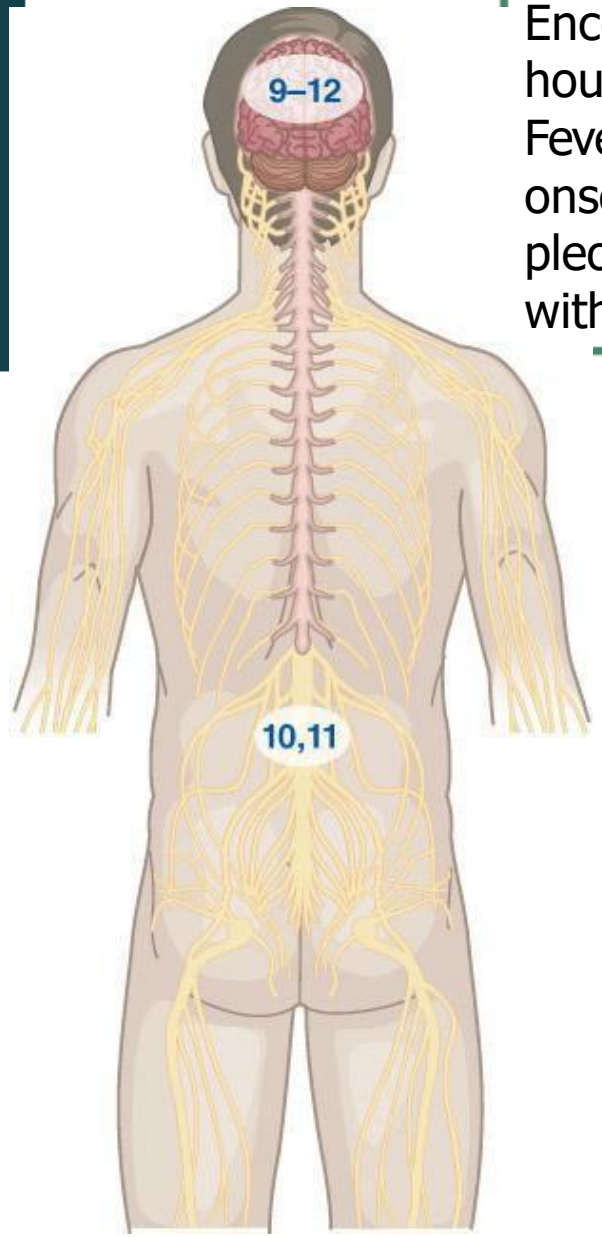
Abscesses – Streptococci, Staphylococci, Toxoplasma gondii, Taenia solium

Empyema – Polymicrobial with Staphylococci and anaerobic gram-negative

Meningoencephalitis – Treponema pallidum, Borrelia burgdorferi, Cryptococcus neoformans, Candida albicans

Symptoms of CNS Infections

Encephalitis – Altered mental state for 24+ hours and 2 (possible) or 3 (probable) of: Fever, headache, new-onset seizures, new-onset focal neurological findings, CSF pleocytosis, abnormal MRI, and EEG consist with encephalitis (gray matter lesions).



Post-infection encephalitis – Similar to infectious encephalitis except mild/no fever and lesions appear in the white-matter.

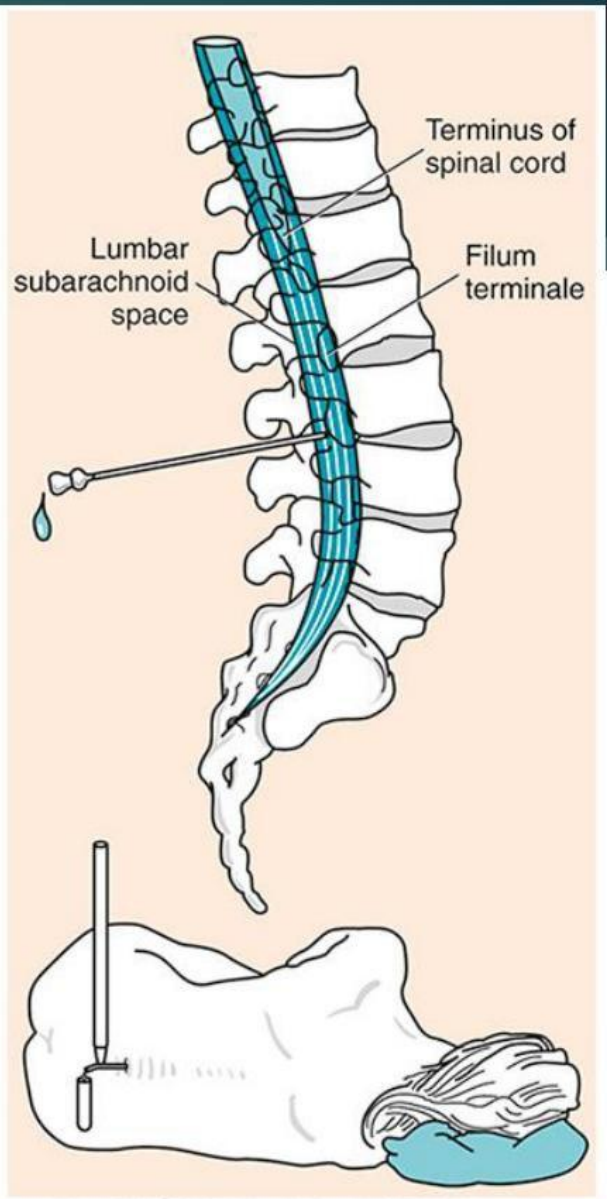
Meningitis – headache, photophobia, nuchal rigidity, fever, and alterations of consciousness in an acute onset

Paralysis – Damage to the peripheral nerves or spinal cord leading to loss of function

Meningoencephalitis – symptoms of both encephalitis and meningitis

CNS viruses

- Gain entry to the CNS via Blood, Macrophages, Neurons (particularly olfactory neurons), or Skin/Muscle-innervating neurons
- Target different areas of the brain/CNS depending on the tropism of the virus OR target specific cell types
- Diagnosed by CSF analysis – increased mononuclear cells (lymphocytes), few to no neutrophils, normal or slightly low levels of glucose
- Confirmed by Ag-specific immunoassay if a particular virus is suspected or by PCR if the cause is unknown
- Season can also contribute to finding a diagnosis



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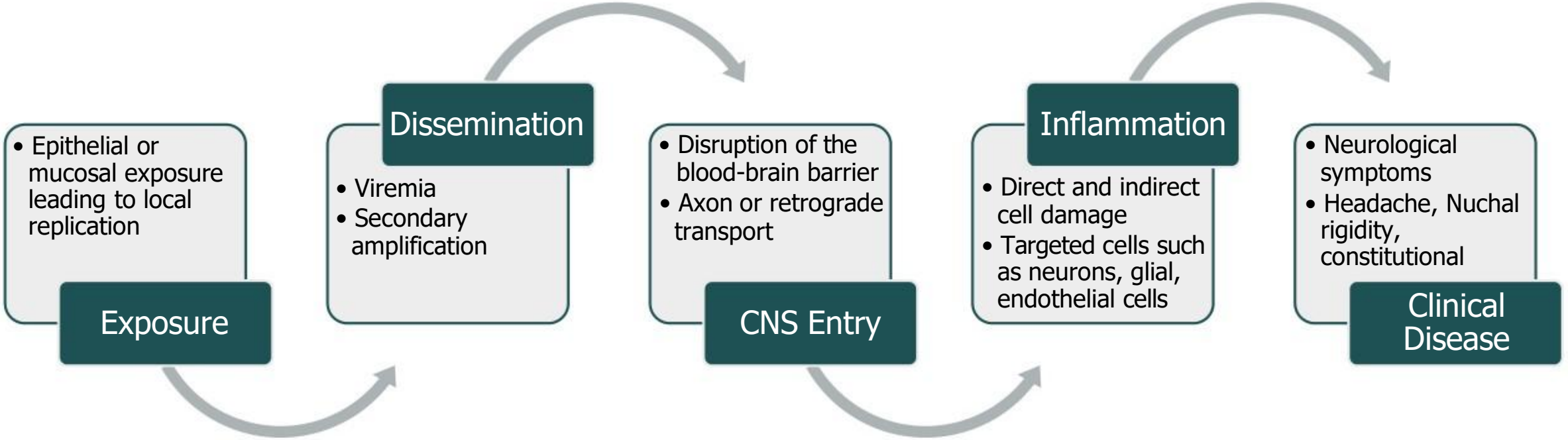
CNS infections – Diagnosis: Bacterial vs. Viral

Key differences between bacterial CSF and viral CSF results:

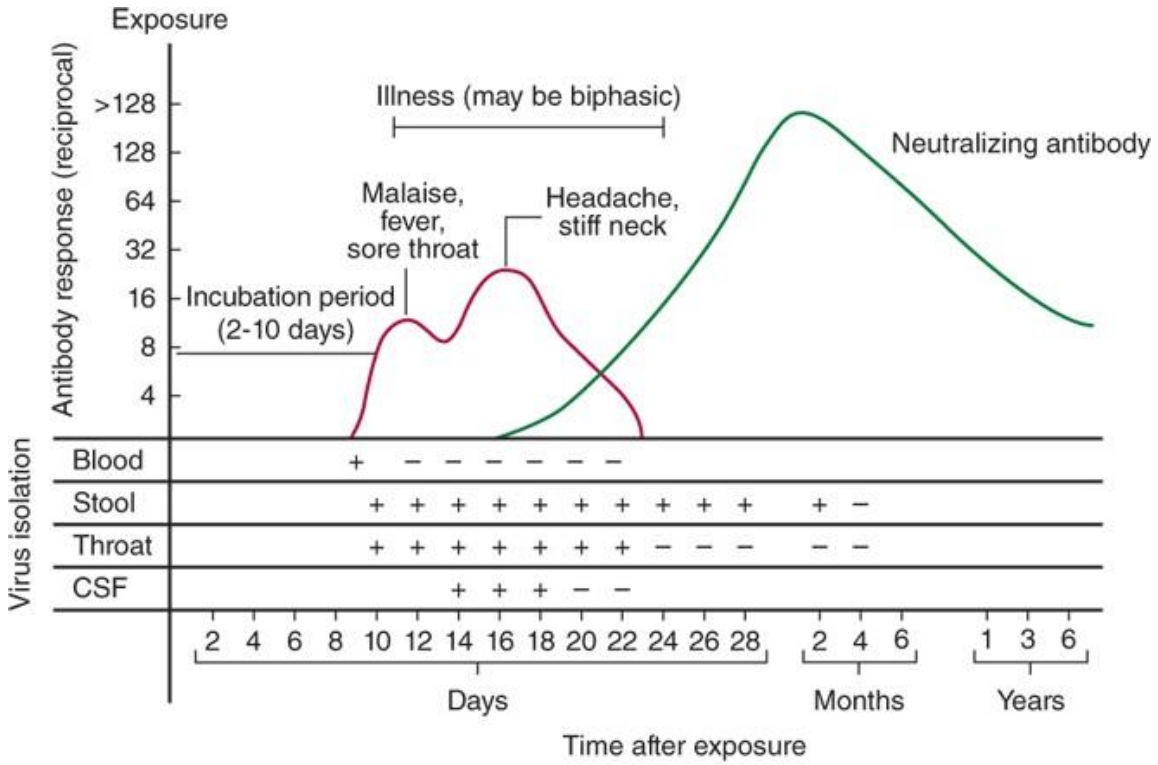
- Organisms in the sample
- Cell type -number
- Protein levels
- Glucose levels

Parameter	Bacterial	Viral
Organisms seen in CSF	Often on Gram stain/culture	None
Cell count (/µL)	100–100,000; PMNs predominate (except Listeria can be mixed/mono)	10–500; lymphocytes predominate (early PMNs possible)
Protein (mg/dL) (10–45 normal)	100–500 (↑)	Normal–mild ↑ (often 50–100)
Glucose (CSF:serum)	Low (<0.4)	Normal
Preferred additional tests	Bacterial antigen test/culture; Multiplex PCR	CSF PCR (enteroviruses, HSV, etc.)

CNS Virus Pathogenesis



Enteroviruses



Source: Kenneth J. Ryan:
Sherris Medical Microbiology, Seventh Edition
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Neurologic Syndromes Associated with Enteroviruses

Syndrome	Virus Type
Aseptic meningitis	Polioviruses 1–3
	Coxsackieviruses A1–11, 14, 16–18, 22, 24
	Coxsackieviruses B1–6
	Echoviruses 1–7, 9, 11–25, 27, 30–33
Encephalitis	Enterovirus 71
	Polioviruses 1–3
	Coxsackieviruses A2, 4–9
	Coxsackieviruses B1–6
	Echoviruses 2–4, 6, 7, 9, 11, 14, 17–19, 25
	Enterovirus 71
	Enterovirus 72 (hepatitis A virus)
Paralytic disease	Poliovirus 1–3
	Coxsackieviruses A2–4, 6–11, 14, 16, 21
	Coxsackieviruses B1–6
	Echoviruses 1–4, 6, 7, 9, 11, 13, 14, 16, 18–20, 30, 31
	Enteroviruses 70, 71
Acute cerebellar ataxia	Polioviruses 1, 3
	Coxsackieviruses A2, 4, 7, 9, B1–6
	Echoviruses 6, 9
	Enterovirus 71
Isolated cranial nerve palsies, especially facial	Poliovirus 1–3
	Coxsackieviruses A10, B5
	Echoviruses 4
	Enteroviruses 70
Chronic infections	Polioviruses 1–3 (vaccine-like strains)
	Coxsackieviruses A15, B3
	Echoviruses 2, 3, 5, 7, 9, 11, 15, 17–19, 22, 24, 25, 27, 29, 30, 33

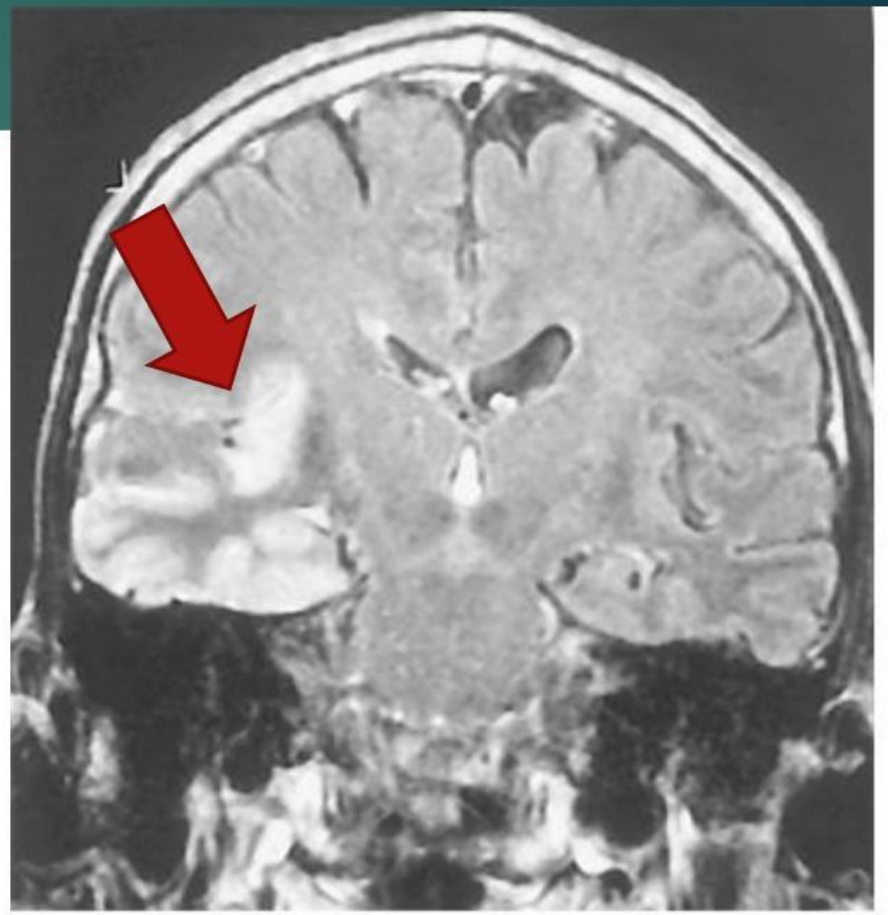
Coxsackieviruses Presentation

- Typically meningitis, but can also cause encephalitis
- Acute onset of fever, chills, headache, photophobia, eye pain, nuchal rigidity – Signs of meningeal irritation
- Nausea and vomiting; Drowsiness or irritability; Febrile illness that comes and goes
- Petechiae, or rash that occurs due to bleeding under the skin and appears in clusters, can occur
  - Glass test – Petechiae will not disappear when pressure is applied like a normal rash would.
- Resolution of symptoms within a week
  - Rare complications include encephalitis, progressive lethargy, disorientation, and seizures
  - Paralysis sporadically seen with coxsackievirus A7 or A9

Syndrome	Coxsackievirus A	Coxsackievirus B
Asymptomatic	+	+
Encephalitis, meningitis, paralytic disease	+ (A7)	+
Carditis	+	+
Neonatal disease	—	+
Pleurodynia	—	+
Herpangina	+	—
Hand-foot-and-mouth disease	+	—
Rash disease	+	+
Acute hemorrhagic conjunctivitis	+ (A24)	
Respiratory tract infections	+	+
Undifferentiated fever	+	+
Diabetes, pancreatitis	—	+
Orchitis	—	+

HSV-1 - presentation

- Febrile illness with symptoms of meningitis AND confusion/behavioral abnormalities, and lethargy/coma,
- Focal or diffuse neurological symptoms – dependent on where the damage is
- Encephalitis – Destruction of one of the temporal lobe causing hemorrhage and erythrocytes in the CSF, seizures, focal neurological abnormalities along with other symptoms of generalized viral encephalitis



Source: J.L. Jameson, A.S. Fauci, D.L. Kasper, S.L. Hauser, D.L. Longo, J. Loscalzo: Harrison's Principles of Internal Medicine, 20th Edition
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Paralysis caused by viral infections

Poliomyelitis

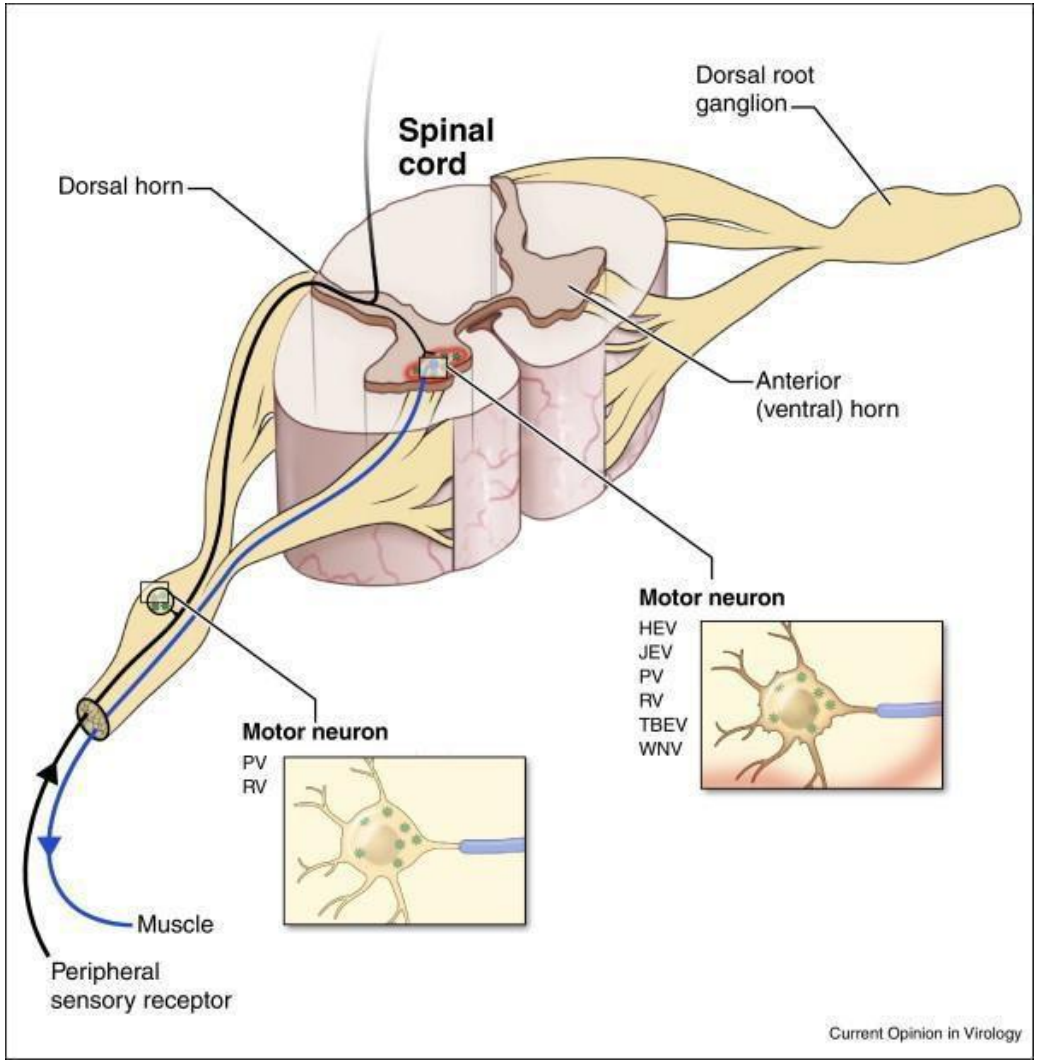
- Flaccid paralysis
- Muscle-wasting
- Hyporeflexia
- Asymmetric

Acute flaccid myelitis

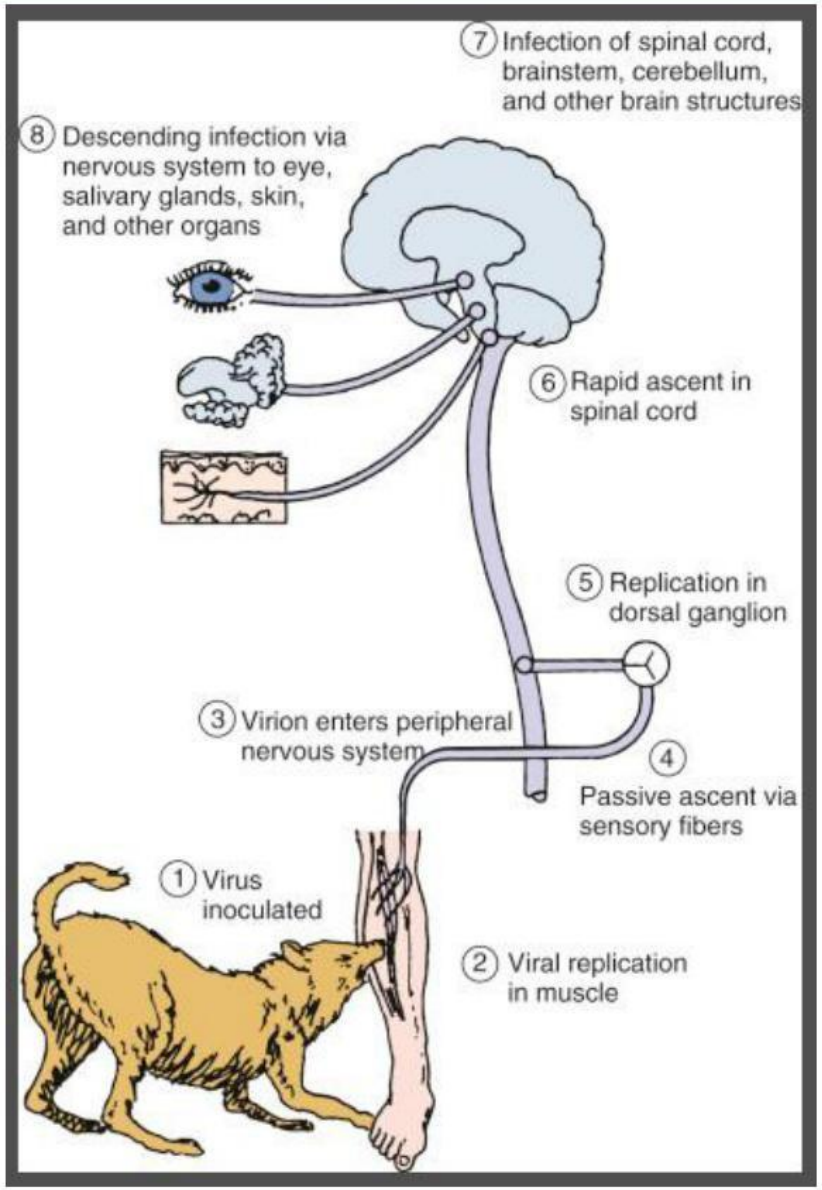
- Flaccid paralysis
- Muscle-wasting
- Hyporeflexia
- Symmetric or asymmetric
- Not caused by poliovirus

Acute transverse myelitis

- Para- or quadriparesis
- Sensory and motor deficits
- Autonomic dysfunction (bladder, bowel, sexual)
- Symmetric



Rabies - pathology



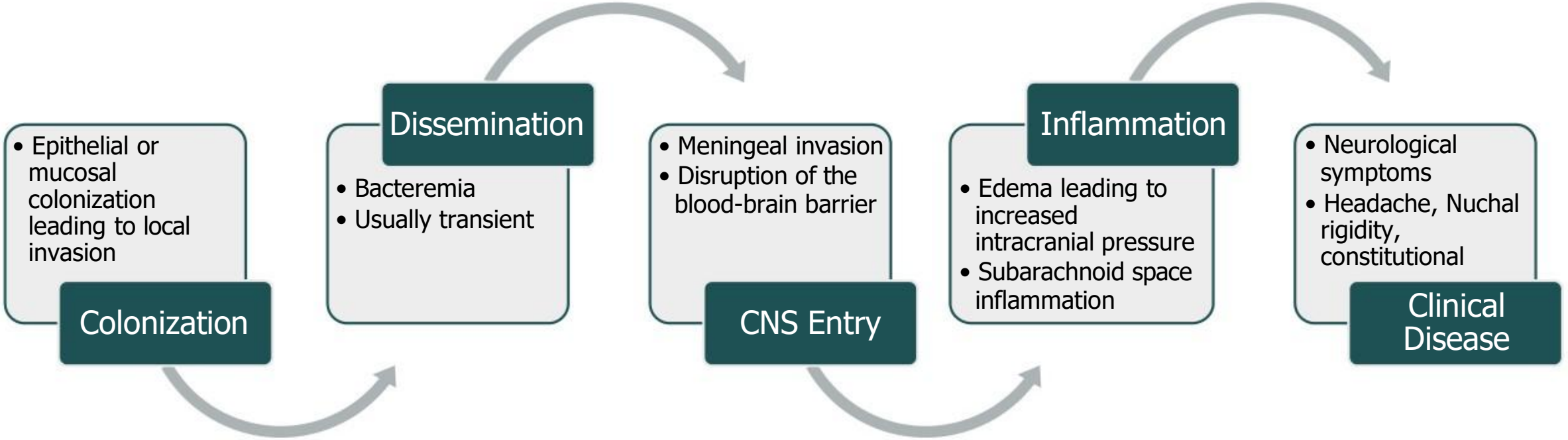
- Replicates in the muscle at the site of bite with minimal or no symptoms (**incubation phase**)
- Incubation time depends on infectious dose and proximity to the CNS/brain
- After weeks or months, virus infects peripheral nerves and travels retrograde to the dorsal root ganglia and the spinal cord to infect the brain (**prodrome phase**)
- Brain infection rapidly leads to neurological symptoms, coma and death (**neurologic phase**)
- Does not elicit antibody response until after the virus has spread to the CNS – early administration of antibody can block progression
- Once clinical disease is apparent, rabies is fatal*
- Long incubation allows for active immunization

Rabies - Presentation

- Prodome phase: Fever, malaise, headache, pain or itching at the site of the bite, GI symptoms, fatigue, and anorexia
- Hydrophobia – characteristic, but only occurs 20 -50% of patients – triggered by pain associated with patient's attempt to swallow water
- Neurological phase: Seizures, disorientation, hallucinations, and paralysis leading to respiratory failure

Disease Phase	Symptoms	Time (Days)	Viral Status	Immunologic Status
Incubation phase	Asymptomatic	60–365 after bite	Low titer, virus in muscle	–
Prodrome phase	Fever, nausea, vomiting, loss of appetite, headache, lethargy, pain at site of bite	2–10	Low titer, virus in CNS and brain	–
Neurologic phase	•Hydrophobia, pharyngeal spasms, hyperactivity, anxiety, depression •CNS symptoms: loss of coordination, paralysis, confusion, delirium	2–7	High titer, virus in brain, salivary gland, and other sites	Detectable antibody in serum and CNS
Coma	Coma, hypotension, hypoventilation, secondary infections, cardiac arrest	0–14	High titer, virus in brain and other sites	–
Death				–

CNS Bacterial Pathogenesis



CNS infections – Diagnosis: Bacterial vs. Viral

Key differences between bacterial CSF and viral CSF results:

- Organisms in the sample
- Cell type
- Protein levels
- Glucose levels

Parameter	Bacterial	Viral
Organisms seen in CSF	Often on Gram stain/culture	None
Cell count (/µL)	100–100,000; PMNs predominate (except Listeria can be mixed/mono)	10–500; lymphocytes predominate (early PMNs possible)
Protein (mg/dL) (10–45 normal)	100–500 (↑)	Normal–mild ↑ (often 50–100)
Glucose (CSF:serum)	Low (<0.4)	Normal
Preferred additional tests	Bacterial antigen test/culture; Multiplex PCR	CSF PCR (enteroviruses, HSV, etc.)

Presentation of acute bacterial meningitis by age

Age group	Symptoms	Signs
Infants (≤2 years)	Irritability	Fever
	Poor feeding	Lethargy
	Vomiting	Stupor, coma
	Unconsciousness	Bulging fontanel
	Respiratory symptoms	Seizures
	Apnea	Petechial or purpuric rash
Children and adults	Headache	Fever
	Neck stiffness or pain	Nuchal rigidity, Kernig's and Brudzinski's signs
	Unconsciousness	Lethargy, confusion, stupor, coma
	Nausea and vomiting	Seizures
	Photophobia	Focal neurologic deficits, including cranial nerve palsies
	Respiratory symptoms	Ataxia (in children)
		Petechial or purpuric rash

CNS infections – Diagnosis: Fungal infections

Parameter	Bacterial	Viral	Tuberculous	Syphilitic (Neurosyphilis)	Fungal
Organisms seen in CSF	Often on Gram stain/culture	None	Rare on AFB smear; culture/NAAT better	None	**Sometimes (India ink/Calcofluor); CrAg preferred**
Cell count (/µL)	100–100,000; PMNs predominate (except Listeria can be mixed/mono)	10–500; lymphocytes predominate (early PMNs possible)	50–500; lymphocytes predominate	10–100 (can be up to a few hundred); lymphocytes	**20–500; lymphocytes predominate**
Protein (mg/dL) (10–45 normal)	100–500 (↑)	Normal–mild ↑ (often 50–100)	↑ (100–500+)	Mild ↑ (≈45–200)	↑ **(≈50–300+)**
Glucose (CSF:serum)	Low (<0.4)	Normal	Low (ratio <0.5)	Normal	**Low (sometimes normal)**
Preferred additional tests	Bacterial antigen test/culture; Multiplex PCR	CSF PCR (enteroviruses, HSV, etc.)	NAAT/PCR, culture;	CSF VDRL (specific, low sensitivity); CSF FTA-ABS/TPPA (more sensitive)	**CSF cryptococcal antigen; India ink/Calcofluor**

Forms of injury with fungal meningoencephalitis

- Chronic meningitis
- Vasculitis
  - Mucormycosis & Aspergillosis
- Parenchymal invasion
  - *Candida* and *Cryptococcus*

Hematogenous spread most common

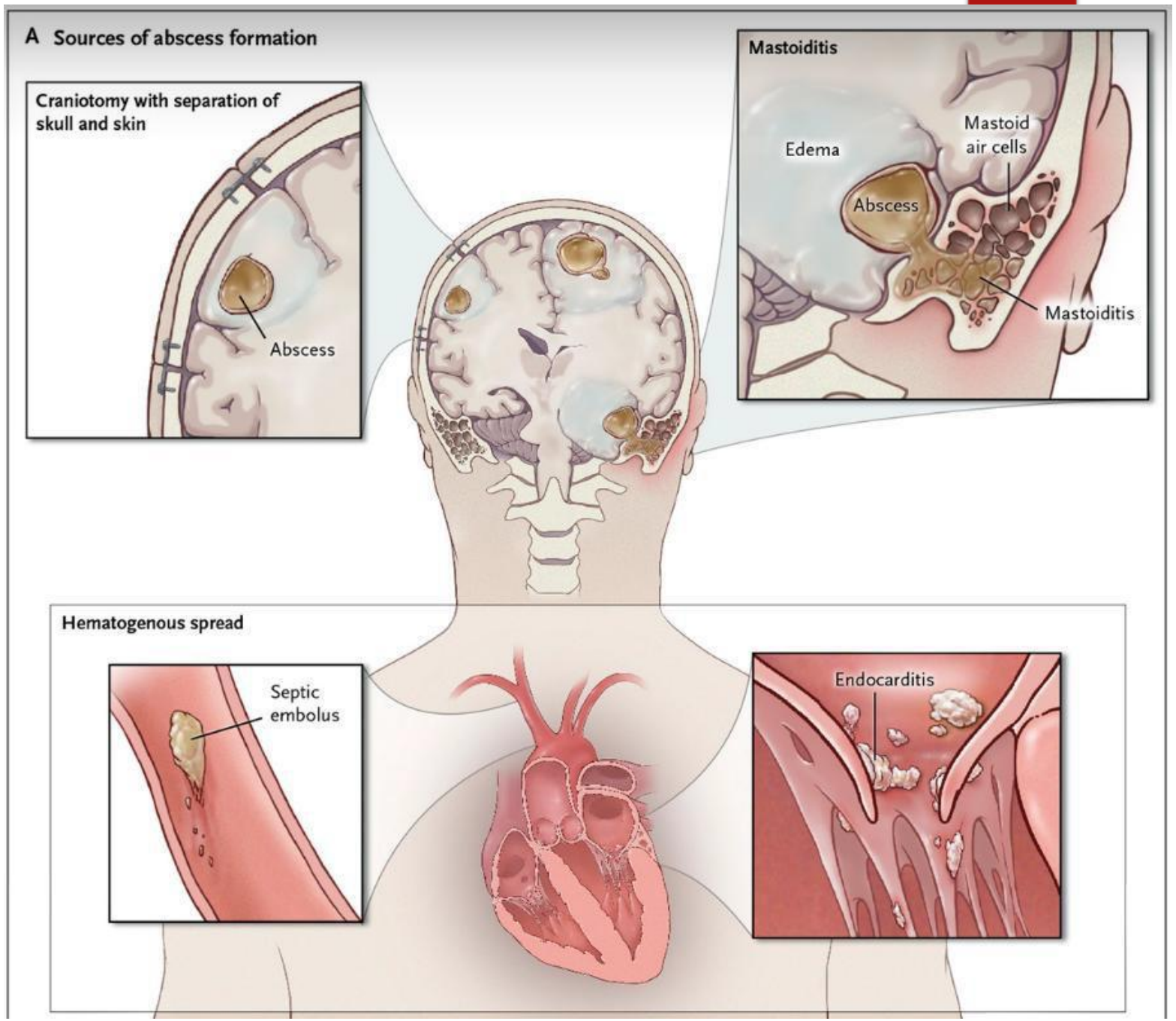
		Chief pathologic manifestations	
Organisms	**Incidence**	**Meningitis**	**Abscess or inflammatory mass**
Cryptococcus	Common	++++	+
Coccidioides	Common	++++	+
Candida	Common	++	++
Aspergillus	Occasional	+	+++
Zygomycetes[b]	Occasional	+	+++
Histoplasma	Occasional	+	+
Blastomyces	Occasional	+	+
Sporothrix	Occasional	+	—

CNS infections – CSF Features

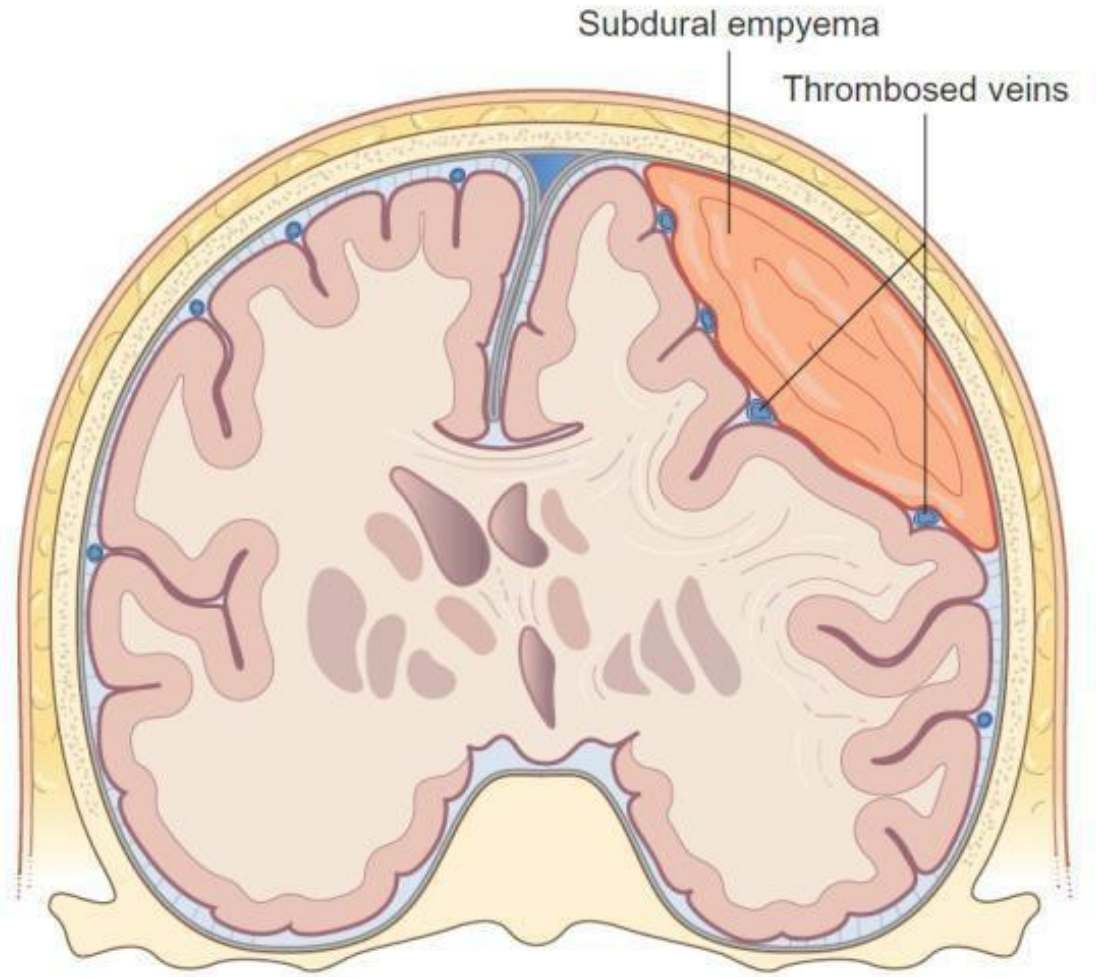
Parameter	Bacterial	Viral	Tuberculous	Syphilitic (Neurosyphilis)	Fungal	Parasitic
Organisms seen in CSF	Often on Gram stain/culture	None	Rare on AFB smear; culture/NAAT better	None	Sometimes (India ink/Calcofluor); CrAg preferred	Rare (except motile amoebas in PAM on wet mount)
Cell count (/µL)	100–100,000; PMNs predominate (except Listeria can be mixed/mono)	10–500; lymphocytes predominate (early PMNs possible)	50–500; lymphocytes predominate	10–100 (can be up to a few hundred); lymphocytes	20–500; lymphocytes predominate	10–500; Eosinophils often predominate. Can be PMNs (Naegleria) or lymphocytes.
Protein (mg/dL) (10–45 normal)	100–500 (↑)	Normal–mild ↑ (often 50–100)	↑ (100–500+)	Mild ↑ (≈45–200)	↑ (≈50–300+)	Mild to moderate ↑ (≈50–200)
Glucose (CSF:serum)	Low (<0.4)	Normal	Low (ratio <0.5)	Normal	Low (sometimes normal)	Normal (can be low, especially in PAM)
Preferred additional tests	Bacterial antigen test/culture; Multiplex PCR	CSF PCR (enteroviruses, HSV, etc.)	NAAT/PCR, culture;	CSF VDRL (specific, low sensitivity); CSF FTA-ABS/TPPA (more sensitive)	CSF cryptococcal antigen; India ink/Calcofluor	CSF differential for eosinophils; Serology; Wet mount; specific NAAT/PCR

Abscess formation

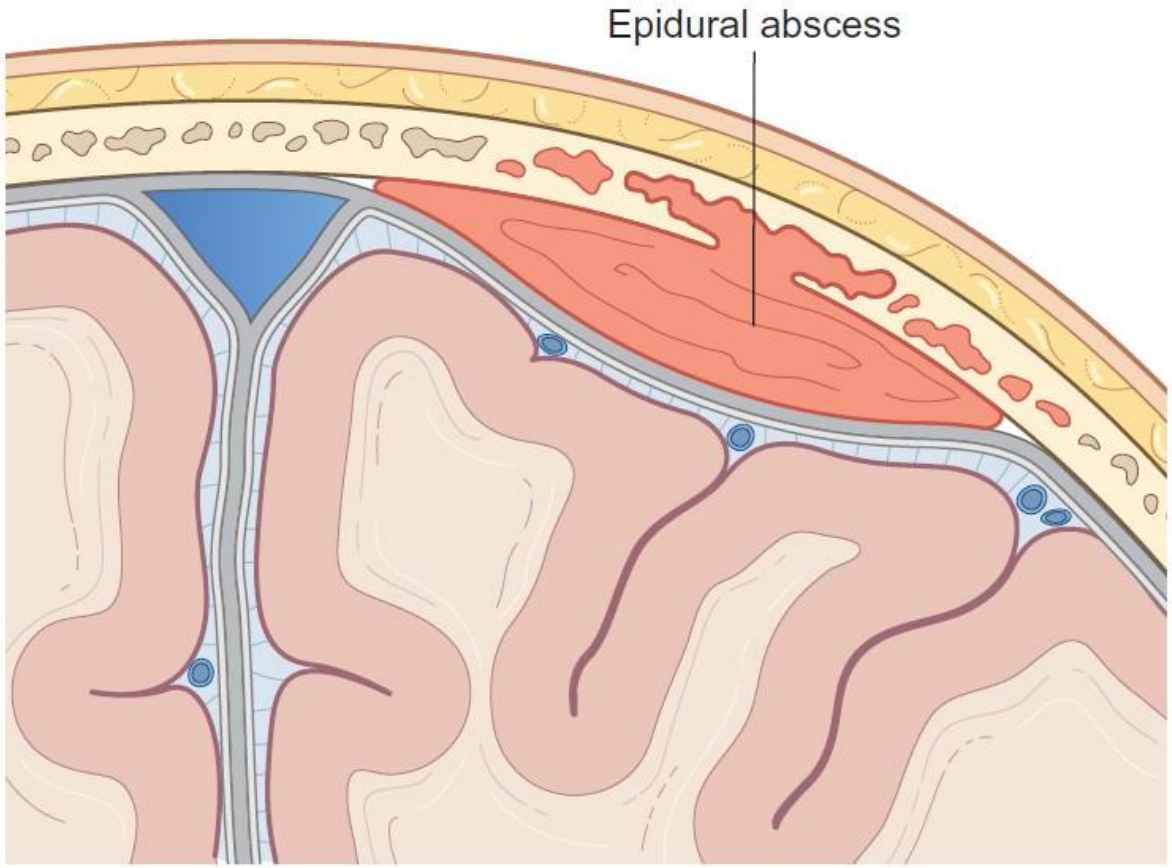
Streptococci, Staphylococci, Toxoplasma gondii, Taenia solium (cysticercosis)



Subdural Empyemas & Epidural Abscesses



Spread from skull or air sinus infections most common



Polymicrobial with Staphylococci and anaerobic gram-negative, or fungal

Major CNS Pathogens

- *Haemophilus influenzae*
- *Neisseria meningitidis*
- *Streptococcus pneumoniae*
- *Listeria monocytogenes*
- *Cryptococcus neoformans*
- *Coccidioides immitus*
- *Toxoplasma gondii*
- *Plasmodium falciparum*
- *Cysticercus cellulosae* (*Taenia solium* larvae)
- *Echinococcus granulosus*
- *Acanthamoeba* sp.
- *Trypanosoma brucei*
- Prions
- Enteroviruses (Echoviruses, Coxsackieviruses, Poliovirus)
- HSV-1
- Varicella-Zoster
- Rabies
- CMV
- Arboencephalitis viruses (West Nile, Western and Eastern equine)